

Experiment No. 7

Symbol Timing Recovery and Synchronization using Polyphase Clock Sync

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Aim

To study symbol timing recovery and synchronization in a QPSK communication system using the Polyphase Clock Synchronization algorithm in GNU Radio.

Objective

The objective of this experiment is to recover the correct symbol sampling instants from a received QPSK signal and achieve synchronization between the transmitter and receiver.

Theory

In digital communication systems, symbols must be sampled at the correct instant for accurate detection. Due to clock mismatches, propagation delays, and channel impairments, the receiver may not know the exact symbol boundaries.

Symbol Timing Recovery is the process of determining the optimal sampling instant for each received symbol. Accurate timing synchronization ensures that symbols are sampled at their peak amplitude, minimizing symbol errors.

A Polyphase Clock Synchronizer is commonly used for symbol timing recovery. It continuously estimates timing errors and adjusts the sampling phase to align the receiver clock with the incoming signal.

The synchronization chain used in this experiment is:

Matched Filter → Symbol Timing Recovery → Carrier Synchronization

Accurate timing recovery improves symbol detection performance and reduces inter-symbol interference.

Equipment Required

- ADALM-PLUTO SDR
- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

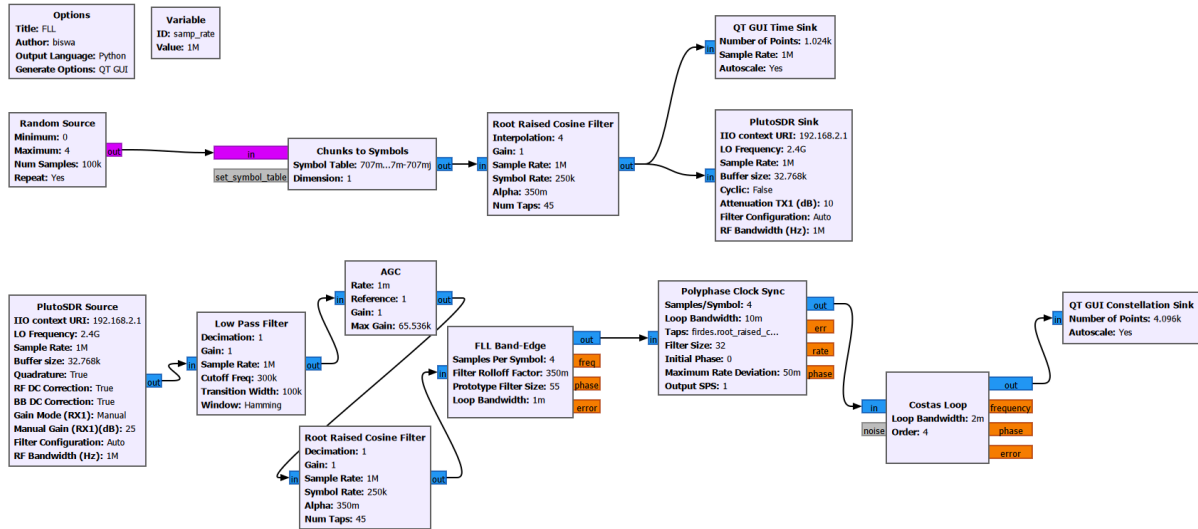


Figure 1: Symbol Timing Recovery using Polyphase Clock Synchronization

Symbol Timing Recovery System

The Polyphase Clock Sync block is highlighted as the primary block responsible for symbol timing recovery and synchronization. Other blocks support signal conditioning and carrier recovery.

Flowgraph Description

PlutoSDR Source

Receives the transmitted QPSK signal.

Low Pass Filter

Removes unwanted frequency components and channel noise.

Automatic Gain Control (AGC)

Normalizes the received signal amplitude.

Root Raised Cosine Filter

Acts as a matched filter and improves signal quality before synchronization.

Polyphase Clock Sync

Performs symbol timing recovery by estimating and correcting timing offsets between transmitter and receiver clocks.

QT GUI Constellation Sink

Displays the recovered constellation points after timing synchronization.

Working Principle

The received signal first passes through a matched filter to improve the signal-to-noise ratio. Due to clock mismatches and transmission delays, the sampling instants may not align with the symbol boundaries.

The Polyphase Clock Sync block continuously estimates timing errors and adjusts the receiver sampling phase. By selecting the correct sampling instants, the receiver successfully recovers transmitted symbols and prepares the signal for carrier synchronization.

Parameters

Parameter	Value
Sample Rate	1 MS/s
Symbol Rate	250 kSymbols/s
Samples per Symbol	4
Roll-off Factor (α)	0.35
Carrier Frequency	2.4 GHz
Polyphase Filter Size	32
Maximum Rate Deviation	0.05
Loop Bandwidth	0.01
Output Samples per Symbol	1

Table 1: Symbol Timing Recovery Parameters

Procedure

1. Open GNU Radio Companion.
2. Create the transmitter flowgraph and configure PlutoSDR transmission.
3. Create the receiver flowgraph using PlutoSDR Source.
4. Add Low Pass Filter and AGC blocks.
5. Add a Root Raised Cosine matched filter.
6. Configure the Polyphase Clock Sync block.
7. Connect a QT GUI Constellation Sink.
8. Run the flowgraph.
9. Observe the constellation before and after timing synchronization.
10. Analyze the improvement in symbol recovery.

Observation

The Polyphase Clock Sync block successfully corrected timing errors and aligned the sampling instants with the received symbols. The recovered constellation became stable and suitable for further carrier synchronization.

Result

Symbol timing recovery was successfully achieved using the Polyphase Clock Synchronization algorithm. Accurate symbol sampling improved synchronization performance and enabled reliable symbol detection.

Applications

- QPSK Communication Systems
- QAM Communication Systems
- Satellite Communication
- Software Defined Radio Receivers
- Wireless Communication Systems
- Digital Modems

Conclusion

The experiment demonstrated symbol timing recovery using Polyphase Clock Synchronization in a QPSK communication system. The synchronization process successfully corrected timing errors and improved overall receiver performance.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. Digital Communication Systems by Simon Haykin
4. Software Defined Radio Principles and Applications